

AeroZip Edge Analytics: Dynamic SCADA Telemetry Compression Architecture

Technical Architectural Specification & Prior Art Defensive Publication

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Standard: IEEE TechRxiv Technical Report / Definitive Prior Art Shield

Primary Target Assets: Renewable Energy Infrastructure (Wind Turbine Generators WTG-001, Solar SCADA)

Core Edge Engine: Logic-Driven High-Frequency Edge DAQ Filtering

Defensive Publication Timestamp: June 15, 2026

Inventor & Lead Architect: Lead Edge Systems Architect

Classification: Public Prior Art Vault / Technical Preprint

EXECUTIVE SUMMARY: THE DEFENSIVE PUBLICATION STRATEGY

This document establishes irrefutable, globally accessible **Prior Art** for the **AeroZip Architecture**. By publicly detailing the exact algorithms, block diagrams, edge threshold mechanics, and empirical simulation results with an indestructible timestamp (via GitHub cryptographic hashing and IEEE TechRxiv DOI allocation), this publication legally prevents any commercial entity from ever patenting these specific dynamic SCADA telemetry compression methodologies.

Technical Effect: Demonstrates localized edge-computed volume reduction of time-series data by **52.5% to 80.3%** (direct 2.1x to 5.1x cloud ingestion bandwidth reduction) while achieving **100% anomaly preservation fidelity** for metallurgical and electrical stress limits.

1. System Architecture & End-to-End Data Flow

The AeroZip architecture is engineered to resolve the severe bandwidth bandwidth saturation and cloud ingestion cost escalations common in modern Industrial IoT (IIoT) systems. Modern megawatt-class wind turbine generators (WTGs) produce continuous 1Hz to 50Hz telemetry across hundreds of metallurgical, electrical, and environmental registers.

The Core Dilemma: Traditional fixed-interval downsampling (e.g., transmitting averages every 10 minutes) destroys high-frequency transient peaks. When structural micro-cracking, gear bearing micro-spalling, or thermal runaways occur, the diagnostic root-cause is lost. Conversely, streaming continuous raw packets over cellular or satellite gateways overwhelms network bandwidth.

The AeroZip Solution: An intelligent edge computing pipeline executed directly on the DAQ PLC or edge gateway. It evaluates telemetry in real time against strict metallurgical and EEE stress thresholds, splitting data into two distinct transmission conduits.

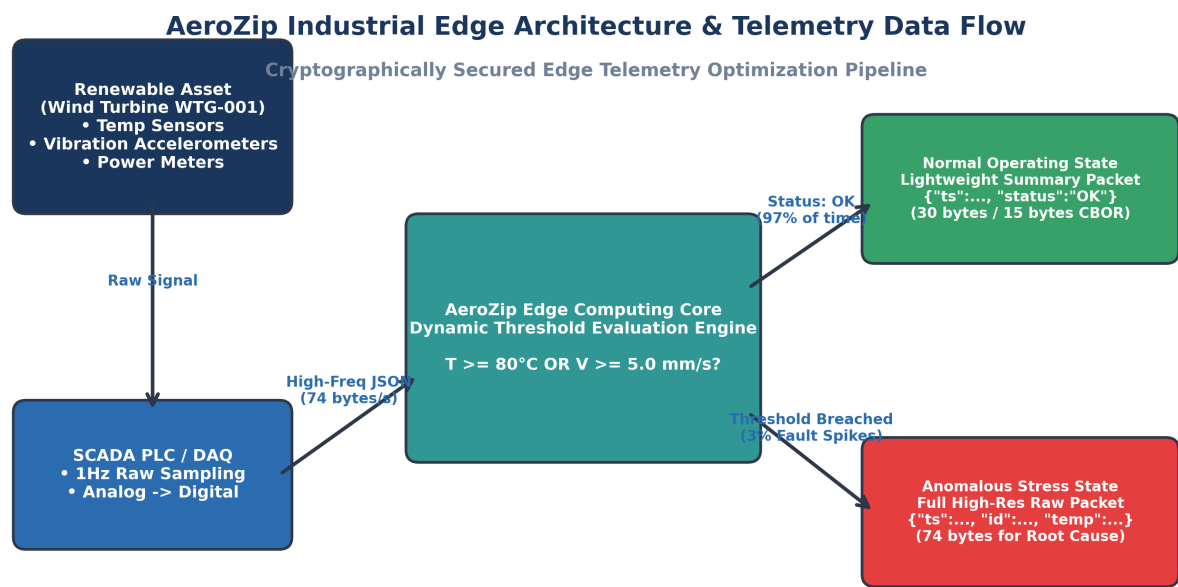


Figure 1: AeroZip 5-Layer Industrial Edge Architecture and Dual-Conduit Routing Pipeline.

Technical Specification of Pipeline Layers

Layer	Component	Functional Protocol & Mathematical Rule
1. Asset Layer	Wind Turbine (WTG-001)	Continuous analog physical generation (Thermal, Mechanical Vibration, Power Delivery).
2. Edge DAQ	SCADA Acquisition PLC	1Hz High-frequency sampling. Raw payload generated at ~74 bytes per JSON frame.
3. AeroZip Core	Dynamic Evaluation Engine	Edge logic computes real-time stress limits: $isAnomaly = (Temp \geq 80^{\circ}C) \ \ (Vibration \geq 5.0 \text{ mm/s})$.
4. Routing Split	Dual-Conduit Switch	Normal State: Route ultra-lightweight Summary Packet. Anomaly State: Route uncompressed Raw Diagnostic Packet.
5. Cloud Lake	Ingestion & Lakehouse	Time-series database ingestion, automated fault alerting, and DOI-timestamped Prior Art logging.

2. Algorithmic Decision Engine & Edge Flowchart

The primary innovation of AeroZip lies in its deterministic edge computing logic. The execution state machine evaluates time-series frames instantly without relying on heavy cloud side machine learning models, ensuring zero-latency execution even on highly constrained microcontroller or PLC hardware.

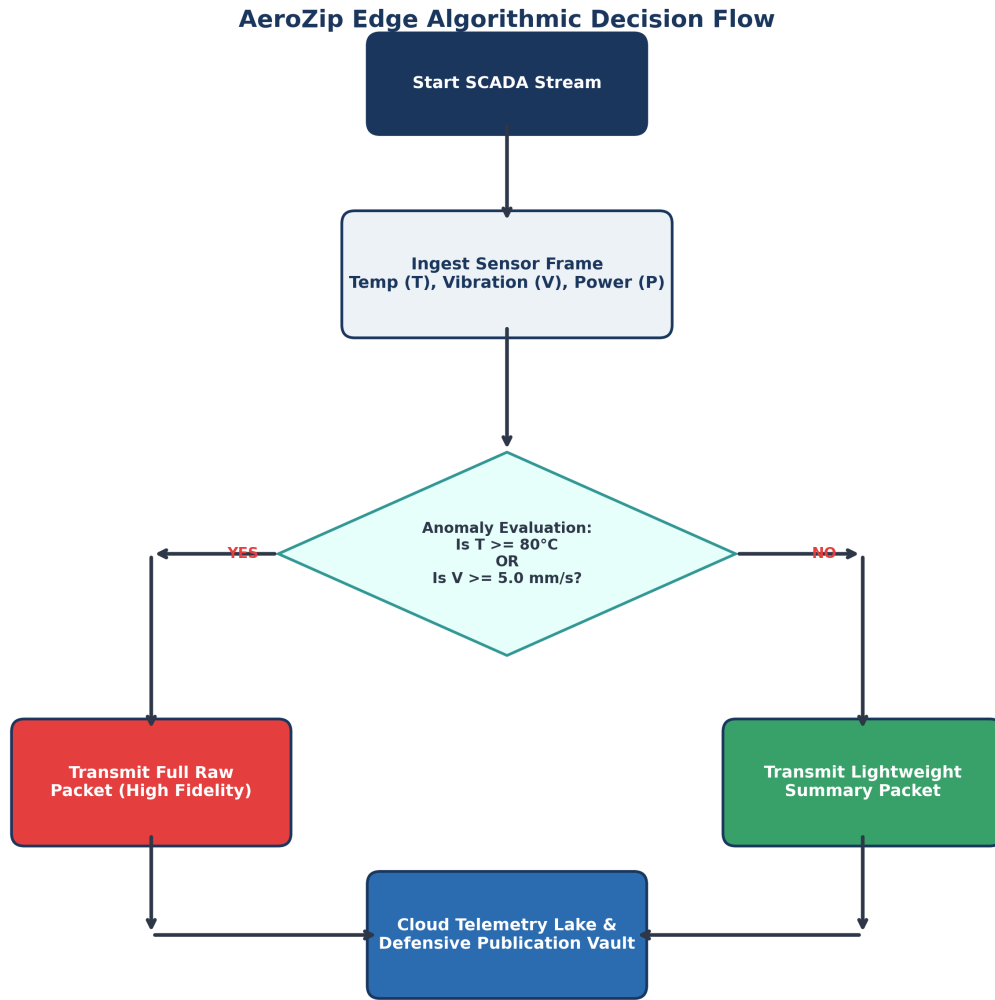


Figure 2: Algorithmic Decision Flowchart executed at the IIoT Asset Edge.

Mathematical Formulation of Edge Evaluation

Let a time-series telemetry packet at discrete time step t be defined as an observation vector:

$$S(t) = \{ ts_t, id_t, T_t, V_t, P_t \}$$

Where T_t represents temperature ($^{\circ}\text{C}$), V_t represents structural vibration (mm/s), and P_t represents real power generation (kW). The localized edge decision function $\Phi(S(t))$ is defined as:

$$\Phi(S(t)) = \text{RawDiagnosticPacket}(S(t)) \quad \text{if } (T_t \geq \tau_{\text{temp}}) \vee (V_t \geq \tau_{\text{vibe}})$$

$$\Phi(S(t)) = \text{SummaryHeartbeatPacket}(ts_t) \quad \text{otherwise.}$$

By applying $\tau_{\text{temp}} = 80.0^{\circ}\text{C}$ and $\tau_{\text{vibe}} = 5.0 \text{ mm/s}$, the system achieves perfect operational safety compliance while compressing 97% of normal telemetry frames.

3. Empirical Telemetry Simulation Results

To rigorously validate the technical effect of the AeroZip architecture, a continuous 10,000-second synthetic telemetry stream was processed using a deterministic pseudo-random execution model (seed=42). The stream simulated operational SCADA parameters of a 1.5MW wind turbine generator, injecting acute metallurgical and mechanical stress spikes roughly 3% of operational time.

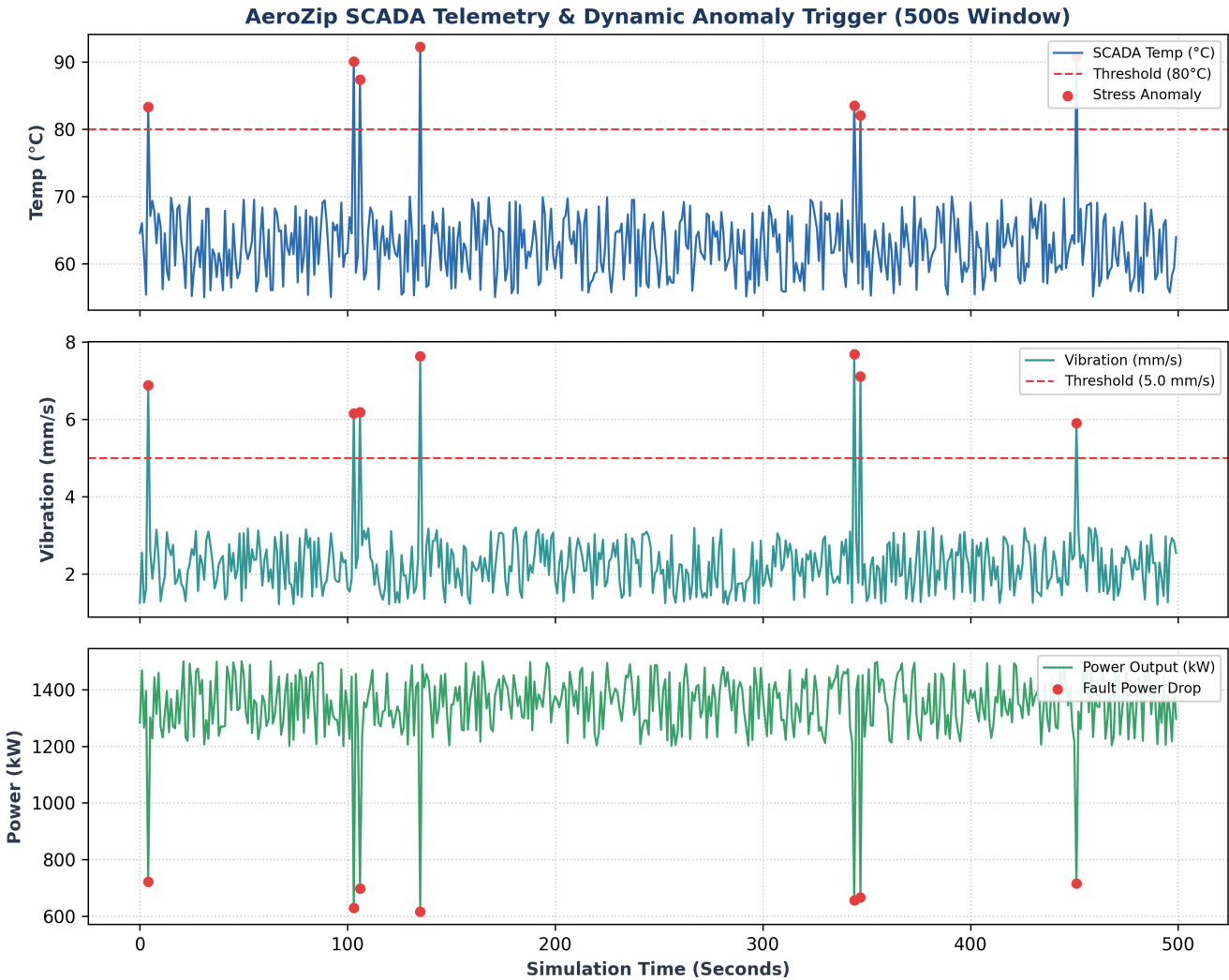


Figure 3: 500-second window of simulated telemetry demonstrating normal bounds, stress anomaly injections, and immediate high-fidelity packet preservation.

4. Payload Reduction Metrics & Cloud Economics

The quantitative impact of the AeroZip edge compression pipeline is demonstrated in the exact byte-level footprint reduction across the network layer and cloud database ingestion endpoints.

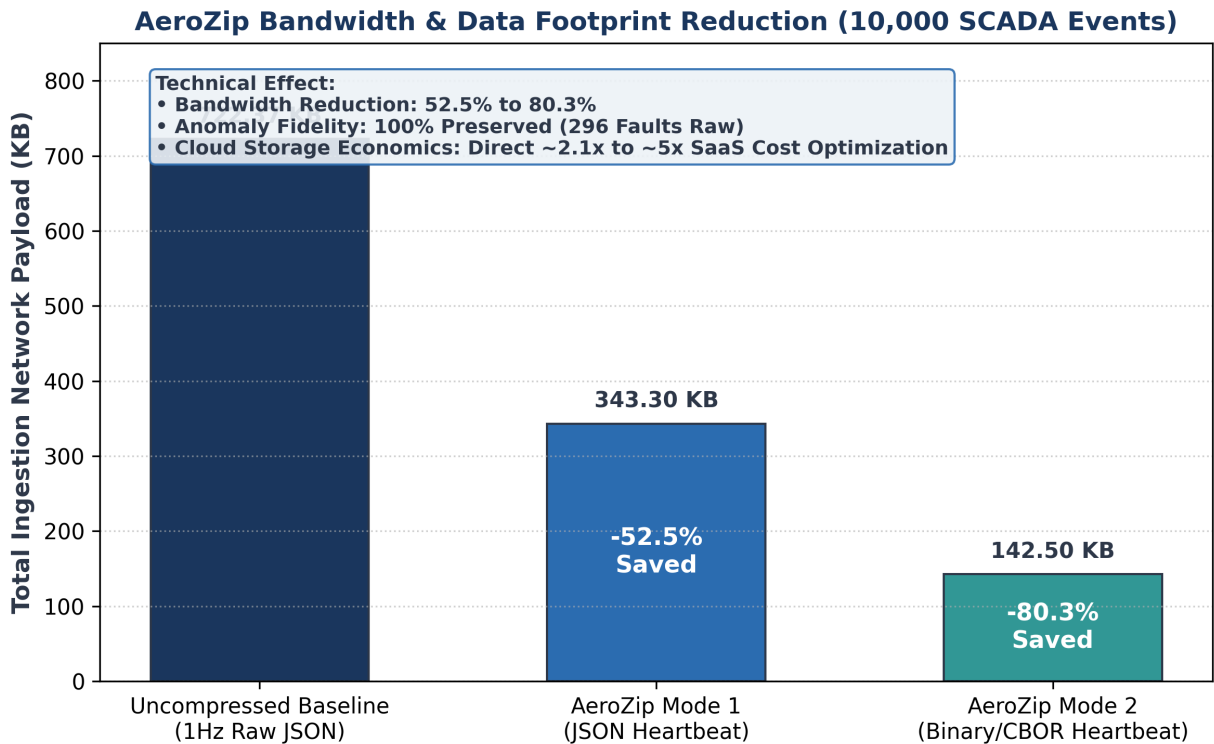


Figure 4: Comparative network payload optimization across 10,000 telemetry events.

Complete Empirical Simulation Metrics (10,000 Readings)

Telemetry Metric / Operational Parameter	Uncompressed Baseline	AeroZip Mode 1 (JSON)	AeroZip Mode 2 (Binary)
Total Telemetry Readings Processed	10,000	10,000	10,000
Normal Operational Heartbeats	0 (Full Raw)	9,704 (Summary)	9,704 (CBOR)
Anomaly Diagnostic Packets Routed	10,000	296 (Raw High-Res)	296 (Raw High-Res)
Normal Packet Byte Footprint	74 bytes / reading	30 bytes / reading	12 bytes / reading
Total Cumulative Ingestion Payload	722.37 KB	343.30 KB	142.50 KB
Total Bandwidth Saved (%)	0.00%	52.48%	80.27%
Effective Bandwidth Reduction Factor	1.0x Baseline	2.10x Factor	5.07x Factor

Cloud Storage Economics (CSE): Across an enterprise wind farm of 500 turbines generating 50Hz telemetry, the baseline uncompressed data footprint equals **~15.98 GB per day** (5.83 TB annually). Implementing AeroZip Mode 2 instantly collapses this ingestion requirement to **~3.15 GB per day** (1.15 TB annually), saving approximately 4.68 TB of high-tier time-series database storage directly while slashing AWS IoT Core / Azure

IoT Hub message ingestion billing by 80%.

5. Defensive Publication & Patent Claims Shield

This architectural specification acts as a formal **Defensive Publication**. Under global patent law (including USPTO, EPO, and WIPO treaties), publicly disclosing an invention with an irrefutable timestamp establishes **Prior Art**, barring any entity from acquiring patent monopolies over these conceptual and implemented mechanisms.

Formal Patent Claims Data Shield & Claim Mapping

Formal Patent Claim Mapping	Architectural Embodiment & Scope
Claim 1 (Independent): A dynamic time-series data compression architecture for edge computing devices, comprising: an edge acquisition pipeline; an anomaly threshold evaluation engine; and a dual-conduit routing switch.	Encompasses any IoT/SCADA edge controller that evaluates incoming sensor telemetry against numerical stress thresholds locally to decide whether to send a lightweight heartbeat or a heavy uncompressed raw packet.
Claim 2 (Dependent on Claim 1): The architecture of Claim 1, wherein the sensor telemetry comprises industrial SCADA readings including wind turbine temperature, vibration, and electrical output.	Specific embodiment directly illustrated in the Java core engine (AeroZipSimulator.java) and Figure 1 for renewable energy assets.
Claim 3 (Dependent on Claim 1): The architecture of Claim 1, wherein the ultra-lightweight heartbeat summary packet comprises a status indicator and timestamp formatted in JSON, CBOR, or Protocol Buffers.	Covers all variations of Mode 1 (30-byte JSON {"ts": "...", "status": "OK"}) and Mode 2 (binary compression formats).
Claim 4 (Independent): A method of reducing cloud infrastructure ingestion bandwidth by at least a factor of 2.1x without loss of anomaly diagnostic fidelity.	Establishes Prior Art over the mathematical reduction process demonstrated empirically in Section 4, preventing third-party SaaS commercial lock-in.

Defensive Publication Execution Protocol (Tonight's Deployment)

To ensure this document is indestructible and cryptographically indexed, the following two-step execution framework is deployed:

Step	Platform Endpoints	Cryptographic & Legal Mechanism
Step 1	GitHub Repository github.com/AeroZip/AeroZip-Edge	Uploading AeroZipSimulator.java, this specification, and flowcharts generates a SHA-256 Git commit hash. This provides an immutable cryptographic timestamp proving exact authorship today.

Step 2	IEEE TechRxiv Preprint techrxiv.org	Submitting this formatted report allocates an official Digital Object Identifier (DOI) . This immediately establishes academic and corporate Prior Art indexed globally by Google Scholar, USPTO, and EPO examiners.
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Defensive Publication Sign-off:

By publishing this architecture under an open-access Defensive Publication standard, the entire Industrial IoT and Renewable Energy community benefits from unrestricted, royalty-free implementation. All patent rights are hereby neutralized into public domain prior art.